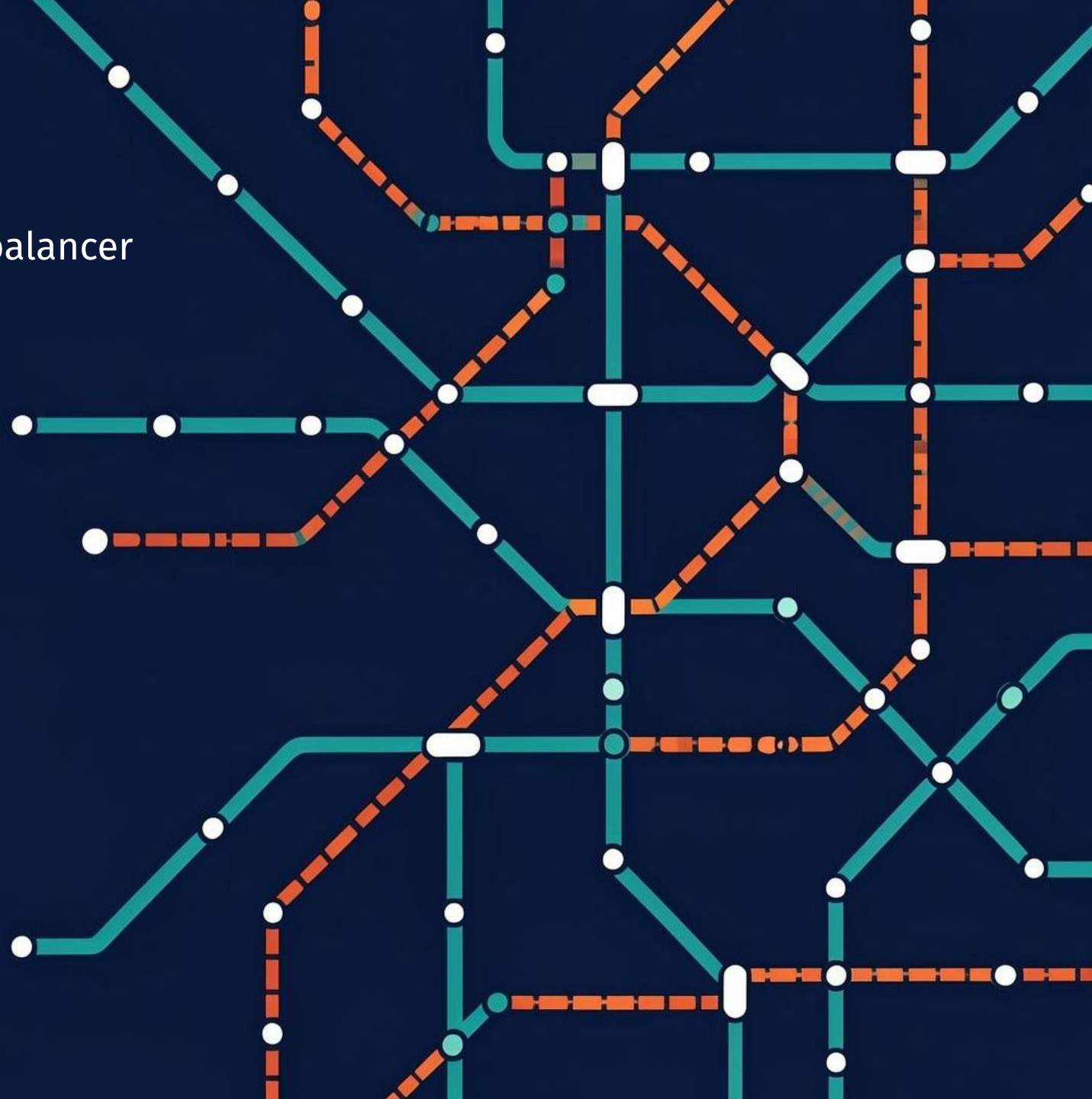
Frustrum

AI/NLP Risk Engine and Tactical Index Rebalancer

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Headlines should update belief, not dictate weight

Markets move on news.

News is noisy, people overreact, and every trade has a cost.

Views stay noisy.

Frustrum turns NLP output into small, bounded, costed trades.

Weights earn their size.

“We do not let a headline set a weight. We let it set a belief, then size the trade by confidence, decay and cost.”

Reader → NLP Risk Engine · Decision maker → Tactical Rebalancer · Scoreboard → Dashboard and backtest



SYSTEM MAP

From news text to explainable, cost-aware trades

A left-to-right path from evidence to bounded action.

News text

Synthetic generated news and social text are the scored inputs.

NLP Risk Engine

FinBERT sentiment · MiniLM event classification · rule-based impact 1–10.

SQLite + FastAPI

Stores views and serves the decision layer.

Rebalancer

Per-stock views, exponential tilt, SLSQP optimisation, and guardrails.

Dashboard + replay

Streamlit dashboard and replay backtest use real daily yfinance prices plus synthetic generated news and social text. A GDELT DOC 2.0 fetcher exists in the repository but is not used by the committed run; no real tweet, Kaggle, NewsAPI, or Reddit data is used.

15-stock mock index · equal-weight benchmark at 6.67% each · AAPL, MSFT, NVDA, GOOGL, AMZN, JPM, BAC, XOM, CVX, JNJ, PFE, WMT, KO, BA, DIS. Engine: ticker linking, novelty/deduplication, and confidence. Real daily prices from yfinance feed correlation and covariance; synthetic generated news and synthetic social text are the scored inputs in the committed pipeline. The GDELT DOC 2.0 fetcher exists in the repository but is not used by the committed run; no real tweet, Kaggle, NewsAPI, or Reddit data is used.

DESIGN CHOICES

Six design choices make the signal usable



Signal fusion

Sentiment × source weight × impact × confidence × novelty × time decay.



Time and novelty

Earnings fade in 1 day; credit events in 10 days. Near-duplicates get novelty 0; contradictory social is halved.



Contagion

Views spill to correlated peers; macro and geopolitical stories spread to sectors.



Optimise

Tilt the benchmark under weight bounds and a 35% sector cap.



Explain

“Why Changed” links moves to headlines.



Prove

Ablations and cost sensitivity test the design. Impact is transparent and rule-based: event base score + sentiment strength + severity keywords.

MEASURED RESULTS

Results are ready for evidence, not invention

Measured backtest results show Frustrum full ahead of the baselines, with materially lower turnover than a naive sentiment tilt.

Measured comparison

Signals ts ≤ bar time · earn the next bar · metrics on the last ~40% after a 60% warm-up · 5 bps on every rebalance, including baselines

STRATEGY	ANN. RETURN	ANN. VOL	SHARPE	MAX DD	ANN. TURNOVER	MEAN IC
Equal-weight hold	19.39%	10.25%	1.8921	-6.92%	0.00%	n/a
Equal-weight daily rebalance	23.79%	10.18%	2.3378	-6.51%	296.50%	n/a
Naive sentiment tilt	23.48%	10.11%	2.3233	-6.55%	1,708.13%	n/a
Frustrum full	**25.61%**	**9.86%**	**2.5960**	**-6.68%**	**749.02%**	**0.0108**

FRUSTRUM ABLATION · SHARPE

No decay **2.4881**
No dedup **2.5888**
No contagion **2.6089**
No cost control **2.5844**

COST SENSITIVITY · SHARPE / ANNUAL RETURN

0 bps · **2.668 / 26.61%**
5 bps · **2.596 / 25.61%**
10 bps · **2.508 / 24.72%**
25 bps · **2.375 / 23.36%**

Synthetic generated news and synthetic social text are the scored inputs; these results validate the end-to-end pipeline, not real-world alpha. The committed run uses no real tweet, Kaggle, NewsAPI, or Reddit data; the GDELT fetcher exists but is not used.

RISK CONTROLS

A desk gets a weak signal with strong guardrails

The system can surface risk without turning one headline into a concentrated bet.

Position limits

Stock weight band 2%–12%; max tilt +/-50% of benchmark; sector cap 35%.

Turnover cap 10% per rebalance · 5 bps transaction cost · 0.5% no-trade band

Emergency logic

Cooldown: 15 minutes.

Circuit breaker: impact >= 9, sentiment <= -0.6, confidence >= 0.5.

Pin the stock at the 2% floor for 60 minutes.

Desk benefit

Less manual news triage, event-driven risk surfaced, explainable allocation changes, transparent limits.

STREAMLIT DASHBOARD · EVIDENCE-BACKED SCHEMATIC

Live Weights · Signal Feed · Why Changed · Performance · NLP Validation · Control

Named tabs mirror the implemented dashboard; no current portfolio weight or screenshot value is asserted.

CONTROL PATH

Signal → constraint → allocation → explanation

FRUSTRUM

Limitations & Next Steps

The boundary between demo and evidence

Assumptions I chose

Half-lives by event type, a sector sensitivity table, an assumed 10 bps return per unit of view, a 5 bps fee, and source trust levels.

Tuned on the training window only.

Data limits

Short sample of large US stocks; English-only FinBERT can misread sarcasm; free feeds have gaps and delays; no real orders or slippage.

Demo news is synthetic. GDELT and tweet/social data are intended real-source next steps—not completed evidence.

Next steps

Fine-tune on labelled finance text; add volume and intraday signals; paper-trade live; add more sources; swap SQLite for a queue to scale.



Weak signal, strong guardrails: here is what we proved and what we didn’t.